

QRX / QubitCoin Whitepaper v0.4

Public Alpha Technical and Economic Specification

This document describes the architecture, tokenomics, and guiding principles of QRX, a decentralized blockchain protocol designed with long-term sustainability and post-quantum readiness in mind.

Disclaimer

QRX is experimental software. The protocol, token model, and network behavior are subject to change. This document is intended for research, testing, and community discussion. No guarantees are provided regarding functionality, value, or rewards.

1. Vision

QRX aims to build a clean, accessible, and future-oriented blockchain system that prioritizes transparency, decentralization, and adaptability. Unlike legacy systems, QRX is designed from the ground up with the expectation that cryptographic standards will evolve over time.

The protocol emphasizes non-custodial participation, allowing users to interact with the network without relinquishing control over their assets.

- 1 Low barrier participation through delegation
- 2 Transparent and predictable emission model
- 3 Future-ready cryptographic design
- 4 Separation of protocol and application layers

2. Architecture

QRX follows a layered architecture that separates concerns between user interaction, consensus, and infrastructure.

- 1 Client Layer: Wallets and applications
- 2 Validator Layer: Nodes participating in consensus
- 3 API Layer: Coordination and communication

3. Cryptography

QRX explores hybrid cryptographic approaches to ensure long-term viability in a post-quantum world.

- 1 Ed25519 for efficient classical signatures
- 2 ML-DSA-65 as a post-quantum candidate
- 3 Hybrid validation model combining both schemes

This approach allows QRX to evolve alongside advancements in cryptography while maintaining backward compatibility.

4. Consensus Model

QRX utilizes a staking-based consensus mechanism where validators are selected based on stake and participation. Delegation enables users to contribute without running infrastructure.

- 1 Delegated staking
- 2 Validator selection by stake weight
- 3 Slashing for misbehavior

5. Tokenomics

Parameter	Value
Max Supply	21,000,000 QUB
Genesis Allocation	2% Dev Fund, 3% Ecosystem Fund, 95% Community
Block Time	10 seconds
Initial Block Reward	0.25 QUB
Halving	Every ~4 years
Network Pool	0% (initial phase)

The emission model is designed to balance early network incentives with long-term sustainability. A lower block reward combined with a relatively fast block time ensures smooth distribution without excessive inflation.

Reward Distribution

- 1 70% allocated to validators
- 2 30% distributed among delegators proportionally
- 3 0% reserved for network pool in initial phase

This distribution ensures that both infrastructure providers and participants are incentivized, while maintaining a transparent and predictable reward flow.

6. Slashing Model

Validators that behave maliciously or fail to meet protocol requirements are subject to penalties.

- 1 Penalty range: 1–5% of stake
- 2 50% of slashed funds are burned
- 3 50% redistributed among active participants

7. Delegation Model

Delegation allows users to assign stake to validators without transferring ownership of their funds. All assets remain in user-controlled wallets, ensuring a non-custodial system.

Rewards are calculated proportionally based on each delegator's share of total delegated stake.

8. Governance

Governance mechanisms are expected to evolve over time. Initial phases rely on off-chain coordination, with future plans for on-chain voting and decentralized decision-making.

9. Roadmap

- 1 Alpha network deployment
- 2 Validator onboarding
- 3 Hybrid cryptography expansion
- 4 Public testing and iteration
- 5 Mainnet evaluation

10. Project Separation

QRX is maintained as an open-source protocol. Application layers, including gamified experiences such as ARES, are developed independently and do not form part of the core blockchain specification.

11. Conclusion

QRX represents a clean and forward-looking blockchain design that prioritizes simplicity, transparency, and adaptability in a rapidly evolving technological landscape.